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Is Acid Rock Drainage (ARD) responsible for any differences in the water chemistry in the area around Mam Tor?

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[1] https://digimap.edina.ac.uk/ (accessed 21st March 2025)

Abstract

The objective of this report is to understand whether water chemistry is affected by Acid Rock Drainage (ARD) around Mam Tor. Field measurements were taken at five localities including Ph, electroconductivity, iron concentration and geological observations. Sites with exposed mudstone and areas that had experienced mass movements were found to have low pH and high iron concentrations, showing an influence of ARD. In comparison, sites with limestone or a less exposed geology had a neutral Ph and lower iron concentrations. Overall, the results clearly indicate ARD has an impact on the water chemistry of Mam Tor, with its local geology being a key factor in its presence among water sources.

Introduction

Acid rock drainage (ARD) describes the contamination of groundwater by acidic waters through elevated amounts of exposed metal sulfides such as iron or pyrite {Jiménez-Castañeda et al., 2016}. It happens as a consequence of the oxidation of sulfides (e.g. pyrite), which generates acidity in the water due to release of H^+ ions and accumulation of toxic metals, sulphuric acid and Fe. {Adams et al., 2007} This process can be associated with active or decommissioned mining sites. Using oxidation of pyrite as an example, it reacts with oxygen and water which forms iron (lll) hydroxides and sulphuric acid as seen below.

$$2FeS_2 + 7O_2 + 2H_2O \rightarrow 2Fe^{2+} + 4SO_4^{2-} + 4H^+$$

Figure 1: An equation showing the oxidation of pyrite

ARD can have severe environmental impacts and even impacts on human health such as reduction of biodiversity and contamination of drinking water {Azapagic et al., 2004}.

In this report I aim to investigate how the underlying geology of a certain area affects the chemistry of the water. In particular, to examine the water chemistry of different areas of Mam Tor and determining whether the physical differences between areas are as a result of ARD. The reason being a past landslide which makes up the majority of the landscape around Mam Tor which may have led to exposing pyrite to oxidation at the surface, as a result leading to ARD-processes which influence the water chemistry of area.

To accomplish this, I will:

- Measure Ph and electroconductivity of water
- Observe geology of each locality
- Use Palin test to identify patterns in water chemistry

Methodology

Data collection will take place in Mam Tor, a large hill in the Peak District in northern England. Here a variety of methods will be used to accomplish the objective.

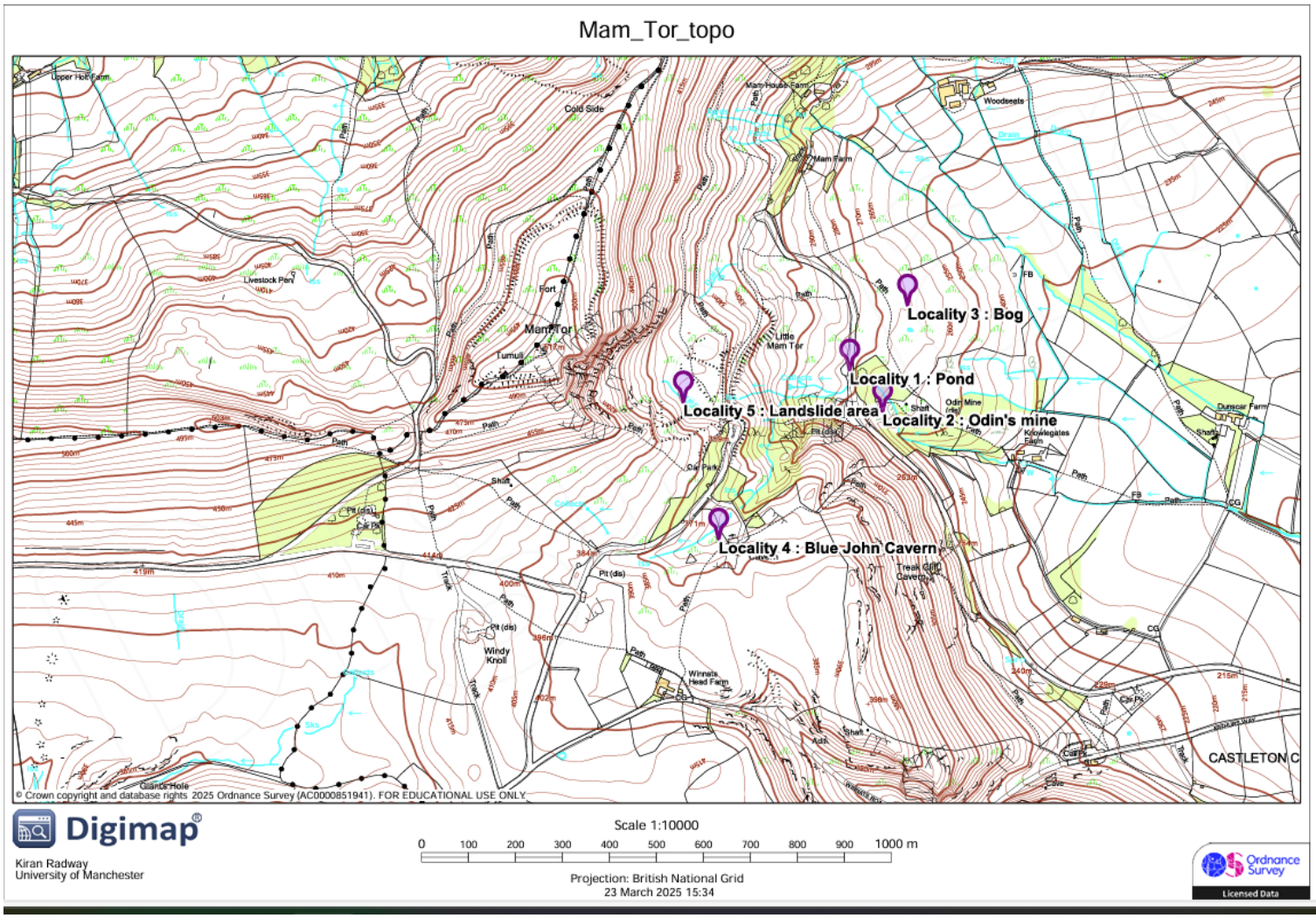


Figure 2 – This is a topographical map of the field area showing all 5 localities

Data collection

Technique	Justification of this technique	Method
Qualitative observations	By using observations, I was able to develop a better idea of the geology and visible differences between localities. This would help to give evidence whether ARD can affect the visible surroundings of an area as a result of altered water chemistry.	In each Locality, observations of the general environment and geology were made in a notebook including rock exposures and surrounding vegetation.
Ph and electroconductivity test	This can be done to show that dissolved H+, sulfate and Fe ions are present in water samples as seen in the equation earlier. Low PH and high electroconductivity are an indication of ARD-affected waters.	A pH probe can be calibrated by placing it in a buffer solution (pH 7). When the reading has stabilised, press the calibrate button. Without calibration readings maybe inaccurate by 1 pH unit Place in a beaker containing a water sample from each locality. A Ph, electroconductivity and temperature measurement are gained from this. The probe must be cleaned every time a new sample is taken.
Fe concentration measurement	Iron ions are products of pyrite oxidation during ARD. The extent of metal sulfide contamination can be connected to ARD. Higher Fe concentration can be linked to low Ph and higher electroconductivity.	A Palintest will be used to measure iron concentration. This is a photometer measuring the absorbance of light as a function of dissolved Fe concentration. A blank sample filled with distilled water is placed first to correct for background absorbance, then a real sample is placed for measurement. Without calibration, results could become offset from the actual value.

Figure 3 – A table showing my methodology and justification behind each method



Figure 4 – This is a Palintest (left)[2] and a PH probe (right)[3] used to measure iron concentration, electroconductivity and Ph

Results

Qualitative data

The descriptions of my localities are written as quoted from my field notebook.

[2] https://acsrm.com/product/palintest-pooltest-9-photometer/ (accessed 28th February 2025)
[3] https://acsrm.com/product/palintest-pooltest-9-photometer/ (accessed 28th February 2025)

Locality 1 - Pond



Figure 5 – This is an image of Locality 1

Description: A pond on the west side of the road next to the gate which blocks off the road, variety of plants in water: lily pads, weeds, low grass, ferns, no rock exposures.

Locality 2 – Odin's mine



Figure 6 – This is an image of locality 2

Description: nearby Odin's mine, a small stream red in colour surrounded by diverse range of vegetation, old mine nearby used for mining copper and lead.

Locality 3 - Bog



Figure 7 – This is an image of Locality 3

Description: very boggy, some plant life, lots of brown ferns, some small rock exposures.

Locality 4 – Blue John Cavern



Figure 8 – This is an image of Locality 4

Description: open field, furthest locality from the landslide, abundant amount of vegetation, no exposed rock

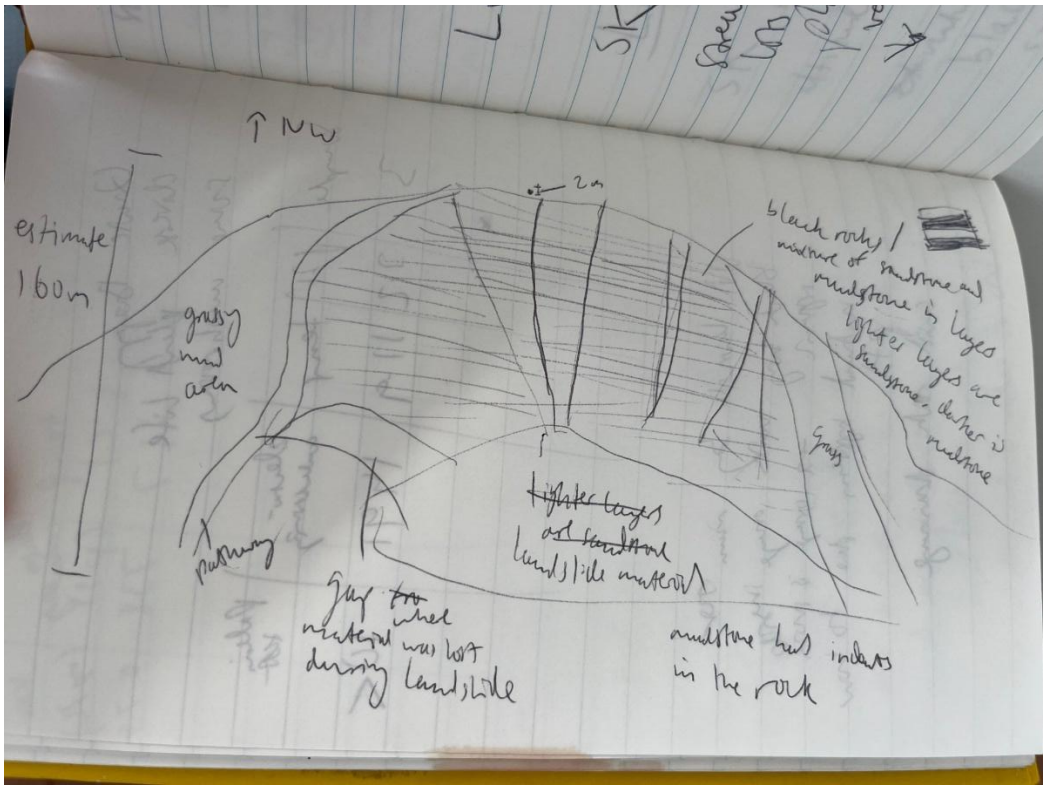


Figure 9 – A sketch made of Mam Tor as observed from Locality 4

A sketch of Mam Tor in the distance where darker mudstone can be seen stacked in rows in between lighter sandstone. Remnants of the landslide are visible too.

Locality 5 – Landslide area



Figure 10 – This is an image of Locality 5

Description: closest area to the source of the landslide, natural exposure of rock, lots of tiny brownish-red rocks and coarser mudstone, some vegetation.

Quantitative data

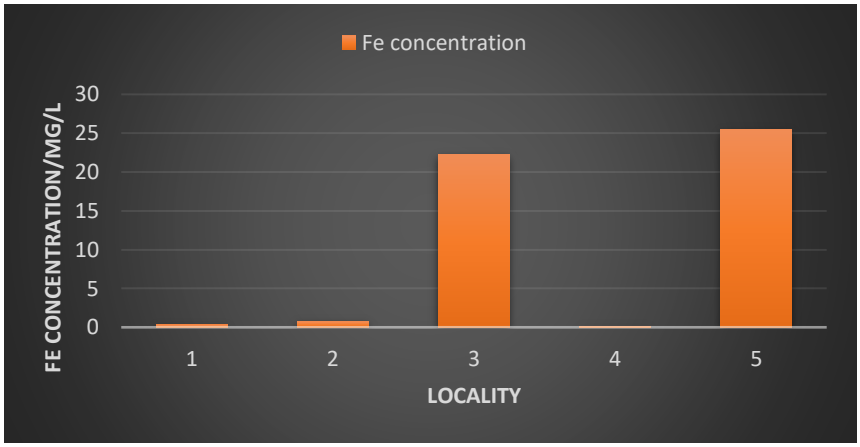


Figure 11 – This is a bar chart iron concentrations of water samples in different localities



Figure 12 – This is a bar chart showing pH of water samples in different localities

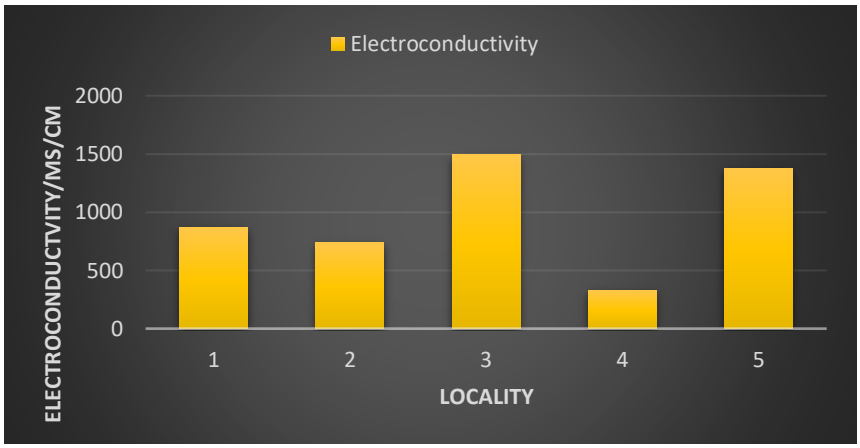


Figure 13 – This is a bar chart showing the electroconductivity of water samples in different localities

Locality	Fe concentration (mg/L)	pH	Temperature (°C)	Electroconductivity (ms/cm)
1	0.34	6	12.6	867
2	0.75	6.4	11.7	738
3	22.2	3.2	11.9	1491
4	0.12	6.2	11.2	330
5	25.5	3	11.8	1377

Figure 14 – A table of recorded measurements from all 5 localities

Discussion

The main aim of this report was to assess whether Acid Rock Drainage causes differences among the water chemistry across Mam Tor. The data collected in the field suggest that ARD is a significant contributor to these variations in water chemistry.

Comparison of graphs

At first, an immediate correlation seen in the graphs (**Figures 9, 10 and 11**) above show a link between low Ph and high electroconductivity and high Fe concentration. This can be seen in localities 3 and 5. This is a typical characteristic of water affected by ARD. The opposite can be seen in localities 1, 2 and 4 showing a limited presence of ARD.

Geological effect

The rock composition of each locality is likely to have an influence in these graph variations. This is seen in sites with exposed mudstone and landslides. By using our results and **Figure 13,** we can see Locality 3 near mudstone and sandstone (MDSS) and locality 5 near mudstone from landslides, both have the ideal conditions for sulfide mineral exposure.

Despite locality 2 being near an old mining site (Odin's mine), it is relatively low compared to locality 3 and 5, but still higher than Locality 1 and 4. The proximity of the mine clearly has some influence on the water chemistry, but not as much as the areas majorly affected by the landslide.

Locality 4 displaying high Ph matches with the limestone it is situated on, showing limestone as a neutraliser of acid. ARD has a minimal effect here.

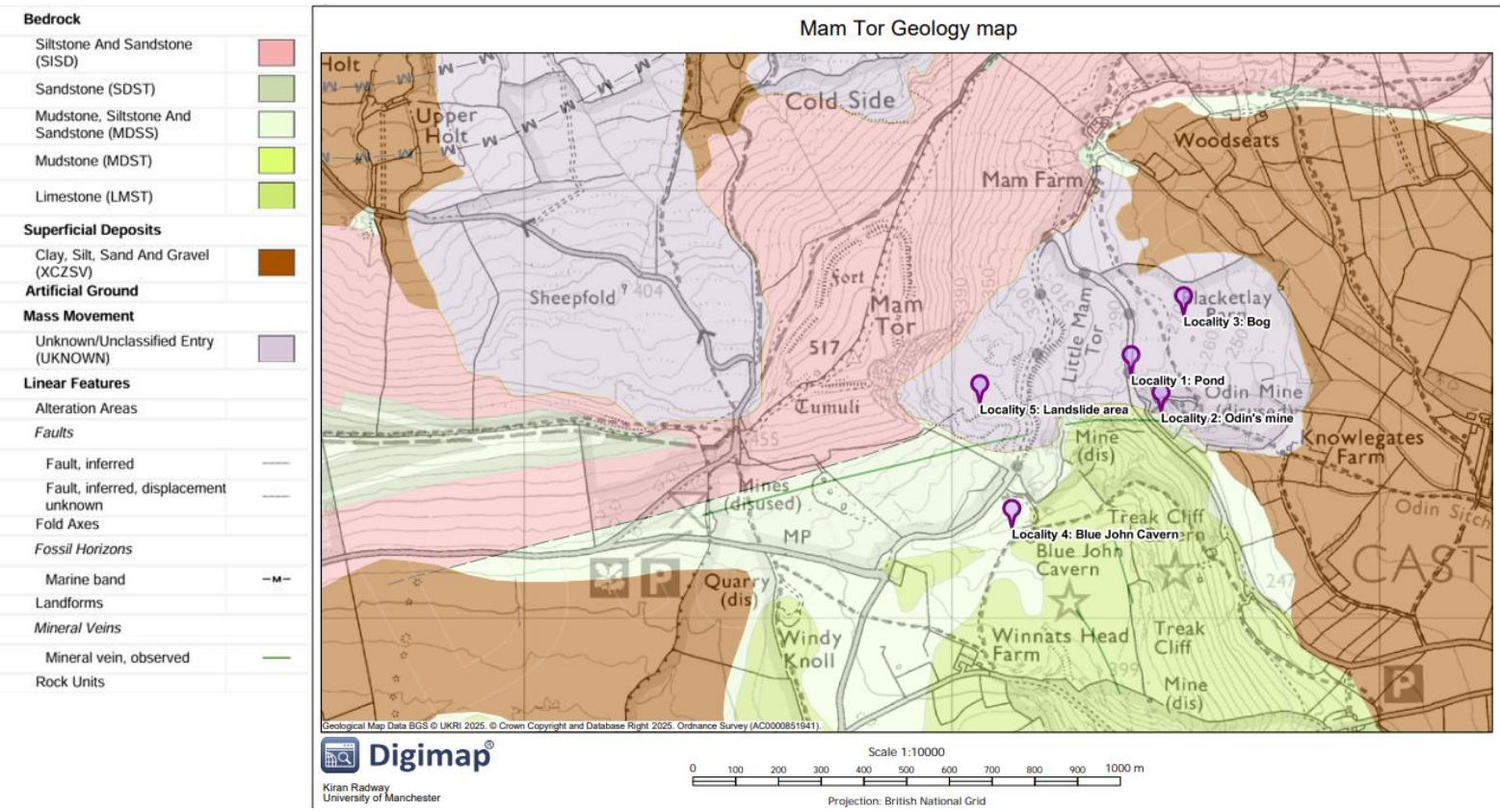


Figure 15 – A geological map of the field area with labelled localities including a legend for rock types

Biological indication

There was also a noticeable correlation between vegetation diversity and water chemistry. Localities 1 and 2 were described to have a diverse plant life whilst having a high pH and lower iron concentrations, showing that ARD can impact biotic conditions within an ecosystem hence showing differences between localities affected by ARD. This aligns with ARD's effect on biodiversity {Azapagic, 2004}.

Furthermore, the Water Framework Directive - United Kingdom Technical Advisory Group (WFD-UKTAG) proposes a threshold Fe concentration of 0.73 mg l-1 total iron as protective of the most sensitive aquatic invertebrate taxa. [4] Localities 2, 3 and 5 exceed this, confirming evidence of ARD-related processes still being active in the area.

[4] https://wfduk.org/sites/default/files/Media/Iron%20-%20UKTAG.pdf (accessed 22nd April)

Uncertainties and limitations

Although links and correlations have been made using the data, there are still several uncertainties that may influence our final results.

- **Accuracy of Fieldwork equipment** – whilst Ph probes and Palintests can be very accurate, they often can be very sensitive to environmental conditions and often need regular calibration for accurate results [5]
- **Sample Limitations** – Only one sample was taken from each locality on the same day. The water chemistry can vary depending on the time of day or weather. Furthermore, samples were only taken from five localities, limiting coverage of the field area. Taking multiple samples from each locality would improve accuracy.

Conclusion

This report shows that Acid Rock Drainage (ARD) is responsible for differences in water chemistry around Mam Tor. Some localities in areas affected by mass movement and containing exposed mudstone are characterised by low Ph and high iron concentration such as locality 3 and 5 indicating ARD presence.

To contrast, localities in areas of limestone abundant ground (Localities 1, 2, 4) showed fewer ARD processes. As a result, they displayed a higher diversity of flora. Although it is important to note Locality 2 did exceed the WFD-UKTAG iron concentration threshold, this may have been affected by the presence of an old mine where excavation led to exposure of metal sulfides. This provides further evidence that geology can be a factor of ARD-driven water chemistry changes.

Overall, this study proves that geology – in particular the exposure of mudstone through mass movements– is the most significant factor in ARD activity. It is also important to note that past mining activity can have an influence too. These factors have led to the release of dissolved ions in the water, increasing acidity and leading to the environmental conditions seen in Mam Tor today.

[5] https://sensorex.com/ph-meters-explained-before-you-buy/#:~:text=The%20main%20advantage%20of%20digital,a%20limitation%20for%20some%20users. (accessed 21st April 2025)

Acknowledgements

I would like to thank Brendan McCormick Kilbride for guiding me throughout all the sections in this report in our tutorial classes and giving me feedback on my first draft.

I would also like to thank the department of Earth and Environmental sciences at the University of Manchester for providing field equipment, transport and guidance for collecting samples in the field.

References

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Appendices

Pictures of raw data taken from my notebook below.

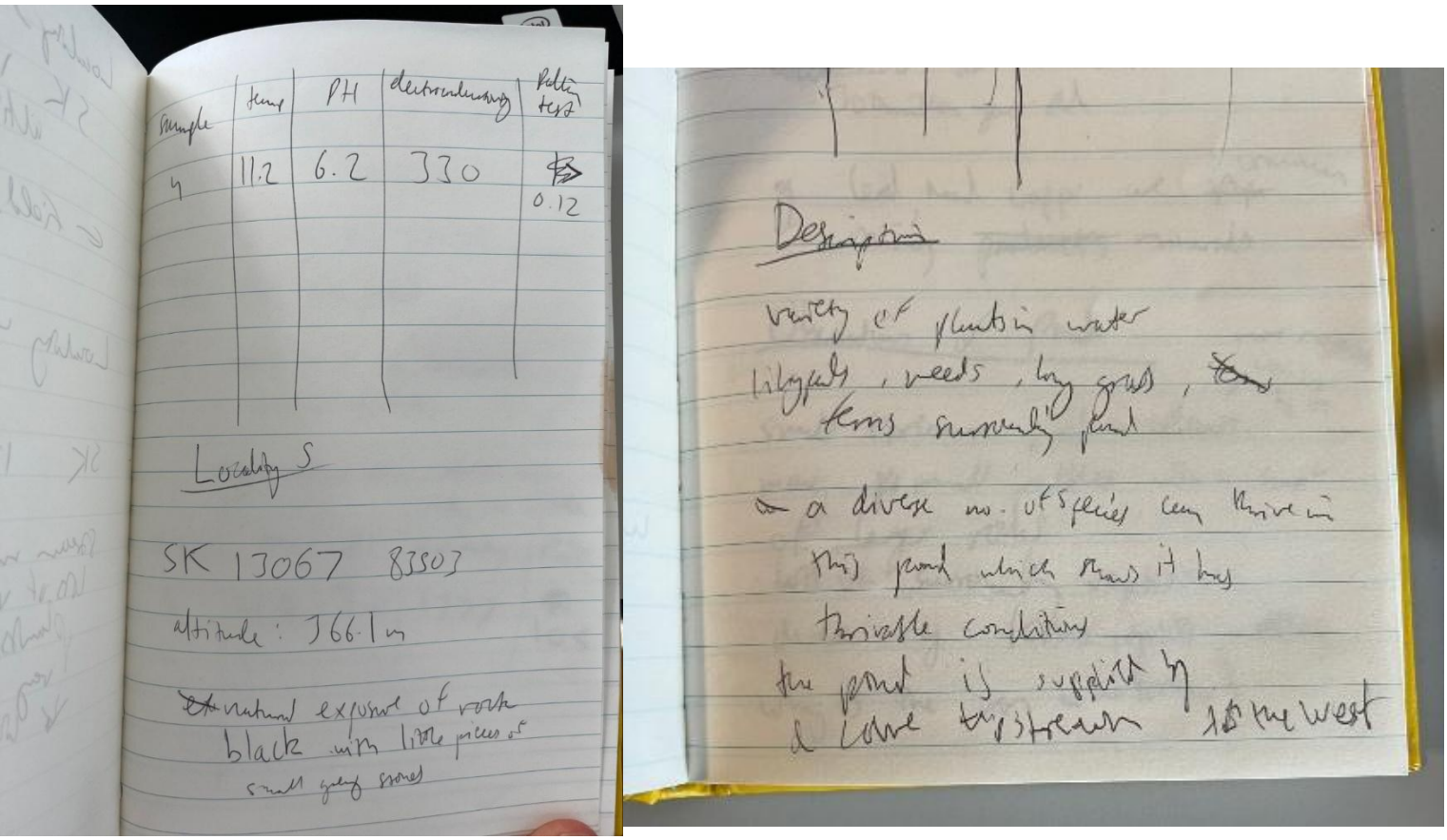


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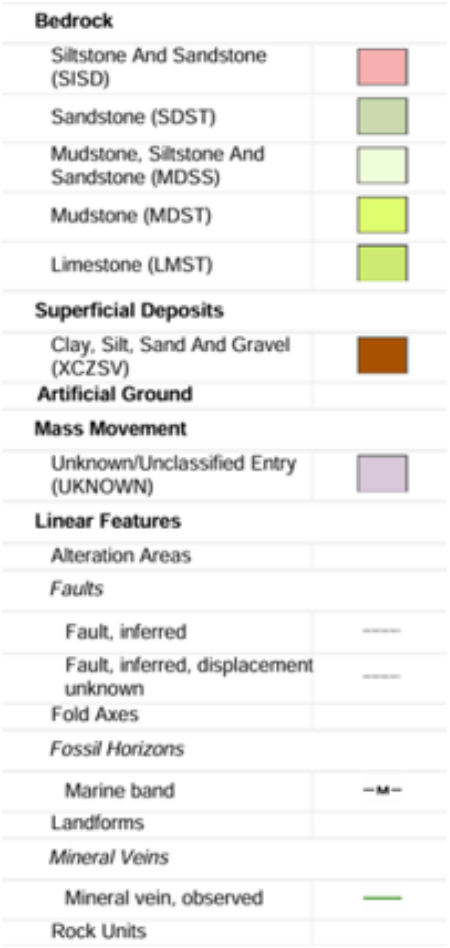


Figure 17 – An enlarged legend for the Geological map (Figure 15)